W2D3

Created time	October 14, 2026 5:13 PM

Q1.

Customer

customer_id
customer_name
customer_city

Order

order_id
order_date
customer_id

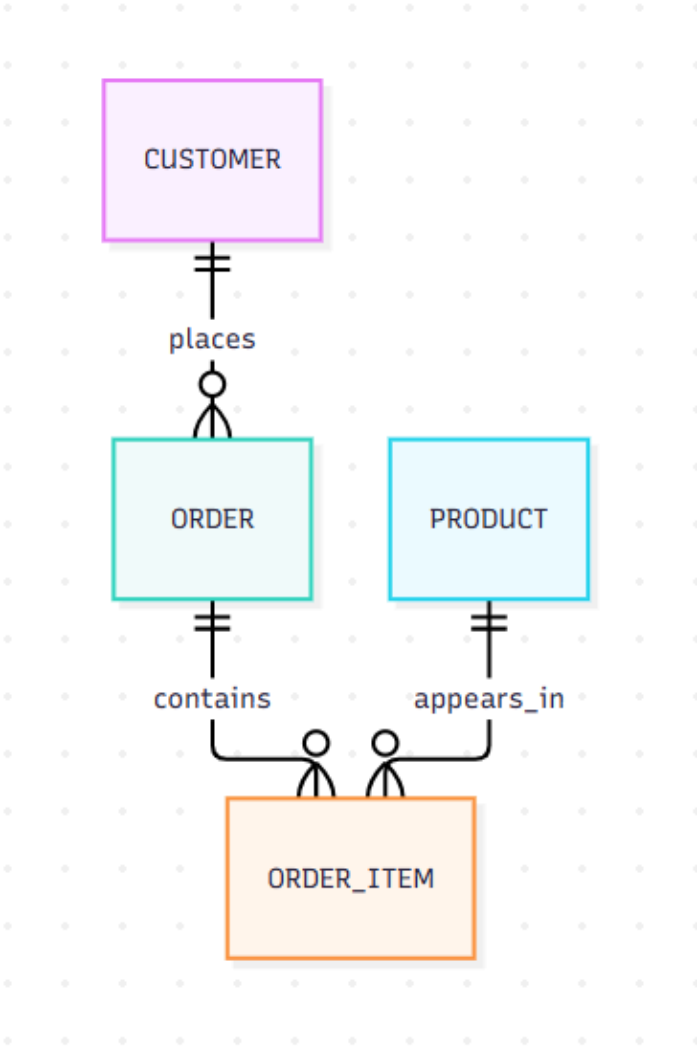
Product

product_id
product_name
unit_price_gbp

Order Item

order_id
product_id
quantity

Q2.



Q3.

Customer : customer_id
customer_name
customer_city

Order : order_id
order_date
customer_id

Product : product_id
product_name
unit_price_gbp

Order_item : order_id
product_id
quantity

Q4. C. Product

Q5.

order_id	customer_id	product_id
101	1	P1
101	1	P2
102	2	P3

Q6.
Products: product_id
product_name

Order_item: order_id
product_id
quantity

Q7 customers: customer_id
customer_name
postcode

postcodes: postcode
city

Q8. B. Reduce repeated facts and keep relationships clear

Q9.

INSERT INTO customers VALUES
(1, 'Amelia Clarke', 'London'),
(2, 'Oliver Bennett', 'Manchester');

INSERT INTO products VALUES
(10, 'Desk Lamp', 25.00),
(11, 'Office Chair', 80.00),
(12, 'Notebook Set', 10.00);

```
INSERT INTO orders VALUES
(1001, 1, '2026-08-01'),
(1002, 2, '2026-08-01'),
(1003, 1, '2026-08-02');

INSERT INTO order_items VALUES
(1001, 10, 2),
(1001, 12, 1),
(1002, 11, 1),
(1003, 12, 3);
```

Q10.

```
SELECT
o.order_id,
o.order_date,
c.customer_name,
c.customer_city,
p.product_name,
oi.quantity,
p.unit_price_gbp,
oi.quantity * p.unit_price_gbp
AS line_total_gbp
FROM orders o
INNER JOIN customers c
ON o.customer_id = c.customer_id
INNER JOIN order_items oi
ON o.order_id = oi.order_id
INNER JOIN products p
ON oi.product_id = p.product_id
ORDER BY o.order_id, p.product_id;
```

Q11.

Model A needs fewer data changes.

Model A: The product name is stored once in the *products* table. Updating *Notebook_set* to *Premium Notebook_set* requires changing only one record.

Model B: The product name is repeated in every order line. Updating the name requires changing every order line containing that product.

Q12. Why might an analytical dataset deliberately repeat some descriptive values?
D. To make some analytical queries simpler to read and use

Q13

FactSales
date_key
customer_key
product_key
store_key
quantity
sales_gbp

Dimensions:
DimDate
DimCustomer
DimProduct
DimStore

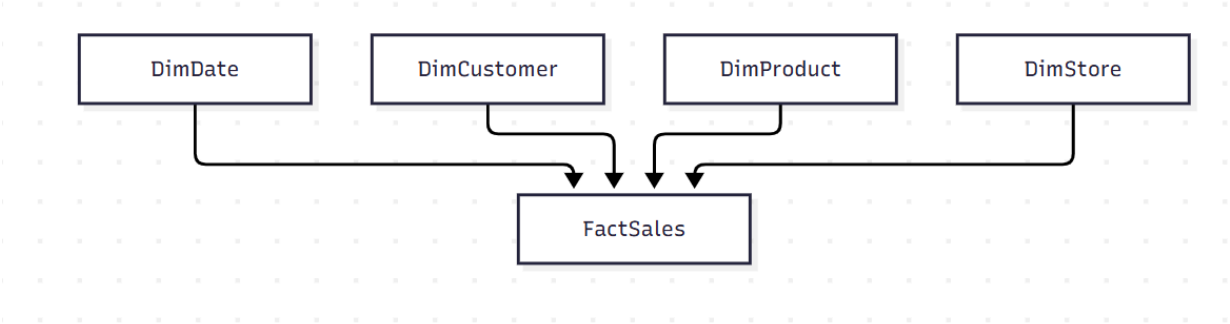
Q14.

Choice B is the better grain.

If one order contains several products, one row per order does not preserve product-level detail.

A clear grain statement is: the grain is **one product line within one order**

Q15



Q16

Which values belong most naturally in a sales fact table?

A. quantity and sales_bgp

Q17.

Customers

customer_id (PK)	customer_name	customer_city
1	Amelia Clarke	London
2	Oliver Bennett	Manchester

Products

product_id (PK)	product_name	unit_price_gbp
10	Desk Lamp	25.00
11	Office Chair	80.00
12	Notebook Set	10.00

Orders

order_id (PK)	order_date	customer_id (FK)
1001	2026-08-01	1
1002	2026-08-01	2

OrderItems

order_id (PK, FK)	product_id (PK, FK)	quantity
1001	10	2
1001	12	1
1002	11	1

PK = Primary Key; FK = Foreign Key.

Q18.

The same customer now appears to have two different cities.

This is an update anomaly caused by repeating the same customer fact across several order rows.

Keeping the customer city in the customer table gives one main place to update it.

Q19. Grain: One row per product line within an order (one order item per product per order).

FactSales; customer_key (FK), product_key (FK), date_key (FK), order_id, quantity , sales_amount_gbp

DimProduct: product_key (PK), name, category

DimDate: date_key (PK), date, month, year

DimCustomer: customer_key (PK), name, city

Q20.

1. What does one row represent?
Define exactly what a single row means, such as one customer, one transaction, or one event.

2. Which fields identify the row?
Identify the primary key or combination of fields that uniquely identifies each record.

3. Which facts repeat?
Find information that appears in multiple records and determine whether it should be stored once and referenced elsewhere.

4. Which fields describe another entity?
Separate attributes that belong to different entities so that each table has a clear purpose.

5. Which relationships must remain valid?
Identify the connections between records and determine which constraints, such as foreign keys, are needed to maintain data integrity.

6. Is the model mainly for transactions or analytics?
Decide whether the system is designed to process day-to-day operations or analyse data for reporting and decision-making.

Why answer these questions first?

Answering these questions before building tables helps establish a clear structure, avoid unnecessary duplication, maintain data accuracy, and ensure relationships are correctly defined. It also helps select an appropriate database design for the intended workload.